Boost converter (12 V)

U1 TPS61085: +5V to 12 V.
FREQ high (1.2 MHz).
EN tied to +5V.
Vout = 1.238 x (1+R80/R81)
= 11.98 V.
R82 33k/C55 1n: comp (TI eq. 9).
C56: soft-start (100n, ~5 ms).
L1 4.7u MWSA0503S (Isat 4.6 A), D22 B5819W.
C2 10u in, C3+C57 out.
Feeds Q2 AO3407A switch.

Keying

TX (MCU PA4) drives Q1 2N7002; R1 22k pull-down holds it off at reset.
Q1 pulls the Q2 AO3407A gate low via R3 22k; P-FET high-side switch for the 12 V PA supply. R2 22k G-S pull-up, C4 10n D-G slows the edges.
R6 ferrite + C7 1u filter.
R8 0.2R 1% PA current shunt; U14 senses it (TP1/TP2).

PA current sense: R8 0.2R shunt, U14 INA193 (gain 20 V/V).
R83/R84 10R + C58 100n differential filter.
R85 1k + C60 100n RC to MCU ADC (PE1).
I_PA = V(TX_ISENSE) / 4.0 V/A.
TP1/TP2 Kelvin test points across R8.

Si5351C synthesizer

XA: U2 25 MHz TCXO, R4 10R feed, R13 OR OE (DNP).
C12 100n to XA; XB open.
J16 CLKIN (10–100 MHz): R43 49.9R termination,
D21 PESD5V0F1BL ESD, C54 100n AC coupling,
R44/R45 10k/10k bias at VDD/2.
OEB tied to GND; outputs enabled over I2C.

Power amplifier

CLK0 (SI5351C) -> U3 SN74LVC1G17 buffer -> C5 100n -> Q3/Q8 BS170 in parallel (DNP, hand-solder), R16/R17 10R gate stoppers.
Drain fed from J3 via L2. R5 3.3k + C8 100n: drain to buffer-out RC. C11 100n DC block to LPF_IN.

RF output: low-pass filter bank (sheet 3)

File: filter_bank.kicad_sch

BAND_A/B/C select one of four LPFs (A B C):
000 80/60 m, 010 40/30 m, 100 20/17 m, 101 15/12/10 m.
J11 BNC output (DNP, hand-soldered).

PA bias

U5 MCP4725 12-bit DAC on SCL_5V / SDA_5V (A0 = GND).
VOUT -> R7 56R -> PA gates.
C9 1u on VOUT, C10 100n VDD.
J1: bias voltage test point.

File: mcu_usb.kicad_sch

File: gps.kicad_sch

File: io_ui.kicad_sch

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RF chain: boost, keying, PA, bias DAC, Si5351C synth, filter bank, BNC out

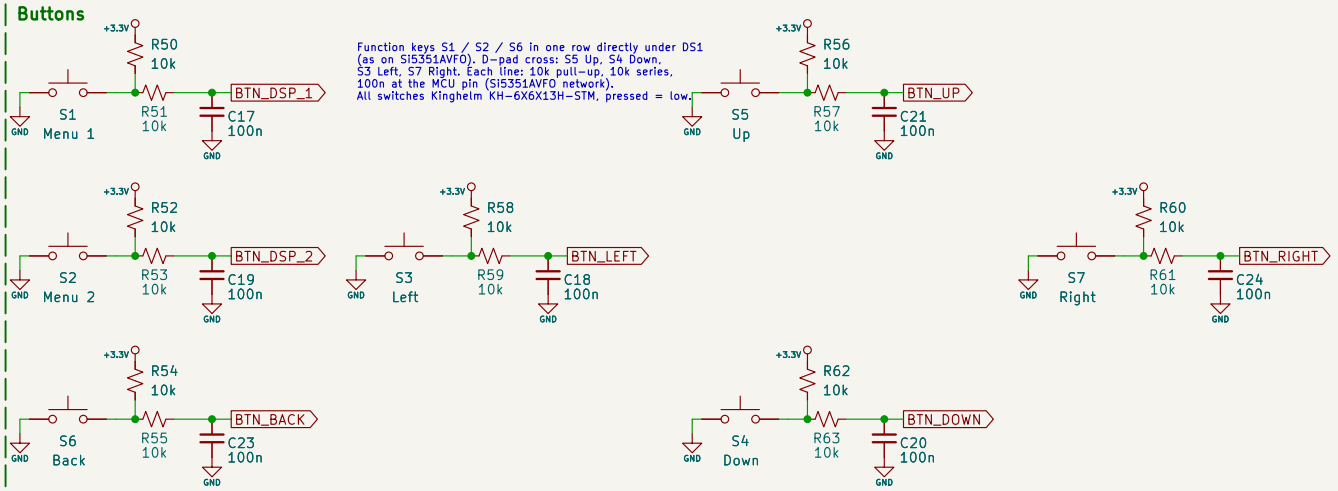
Etherkit

Sheet: /
File: OpenBeacon2.kicad_sch

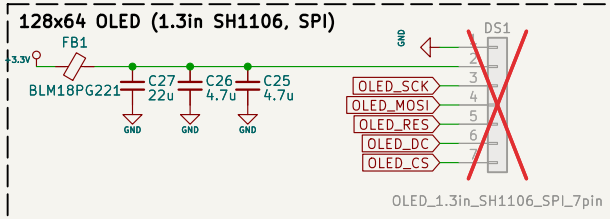
Title: OpenBeacon 2

Size: A4	Date: 2026-10-25	Rev: C
KiCad E.D.A. 10.0.7		Id: 1/6

Buttons



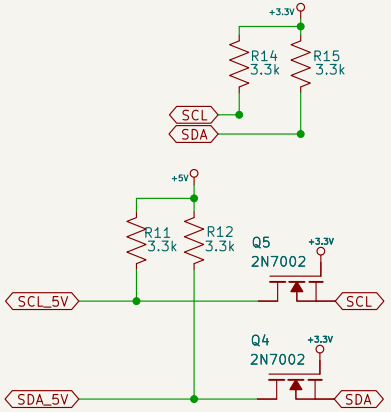
128x64 OLED (1.3in SH1106, SPI)



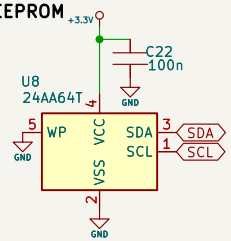
Same module and pinout as SI5351AVFO DS1: DNP, kit-supplied, user-soldered, 3.3 V logic, no level shift.
FB1 + C25-C27 copy the VFO OLED rail filter at the header.
SPI1 ALT1: SCK PC6, MOSI PC4; CS PC7, DC PB4, RES PB5.

I2C level conversion

Q4/Q5 2N7002 bidirectional shifters: SDA/SCL (3.3 V, R14/R15 3.3k) to SDA_5V / SCL_5V (R11/R12 3.3k to +5V) for U5 MCP4725 on sheet 1.



External EEPROM



U8 24AA64T 64 Kbit I2C EEPROM on the 3.3 V SDA/SCL bus.
WP tied to GND (writes enabled).
C22 100n decoupling.

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I/O headers, SH1106 SPI OLED, 7 buttons, I2C level shift, 24AA64

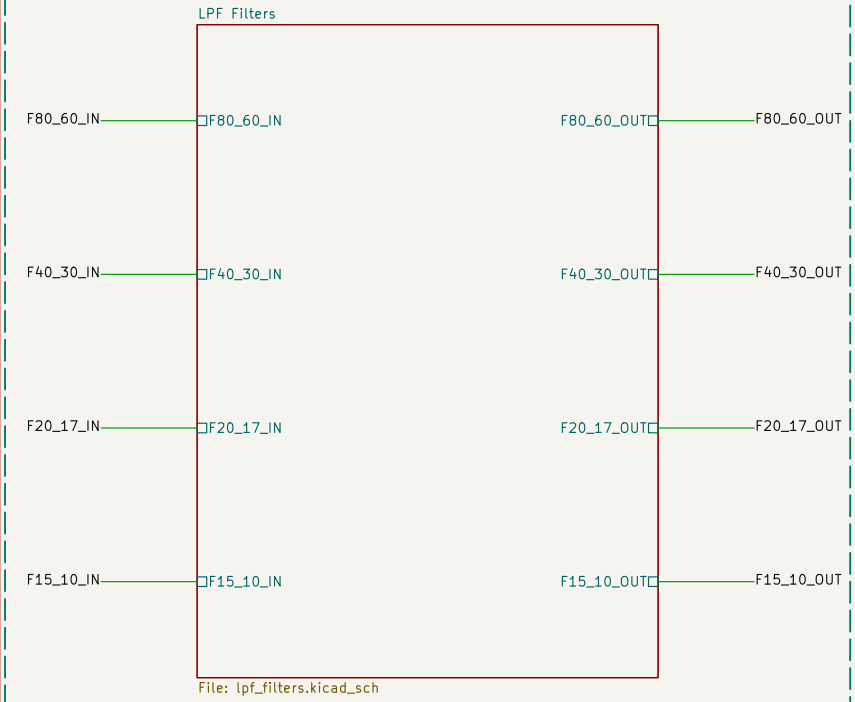
Etherkit

Sheet: /I/O + UI/
File: io_ui.kicad_sch

Title: OpenBeacon 2 – I/O and User Interface

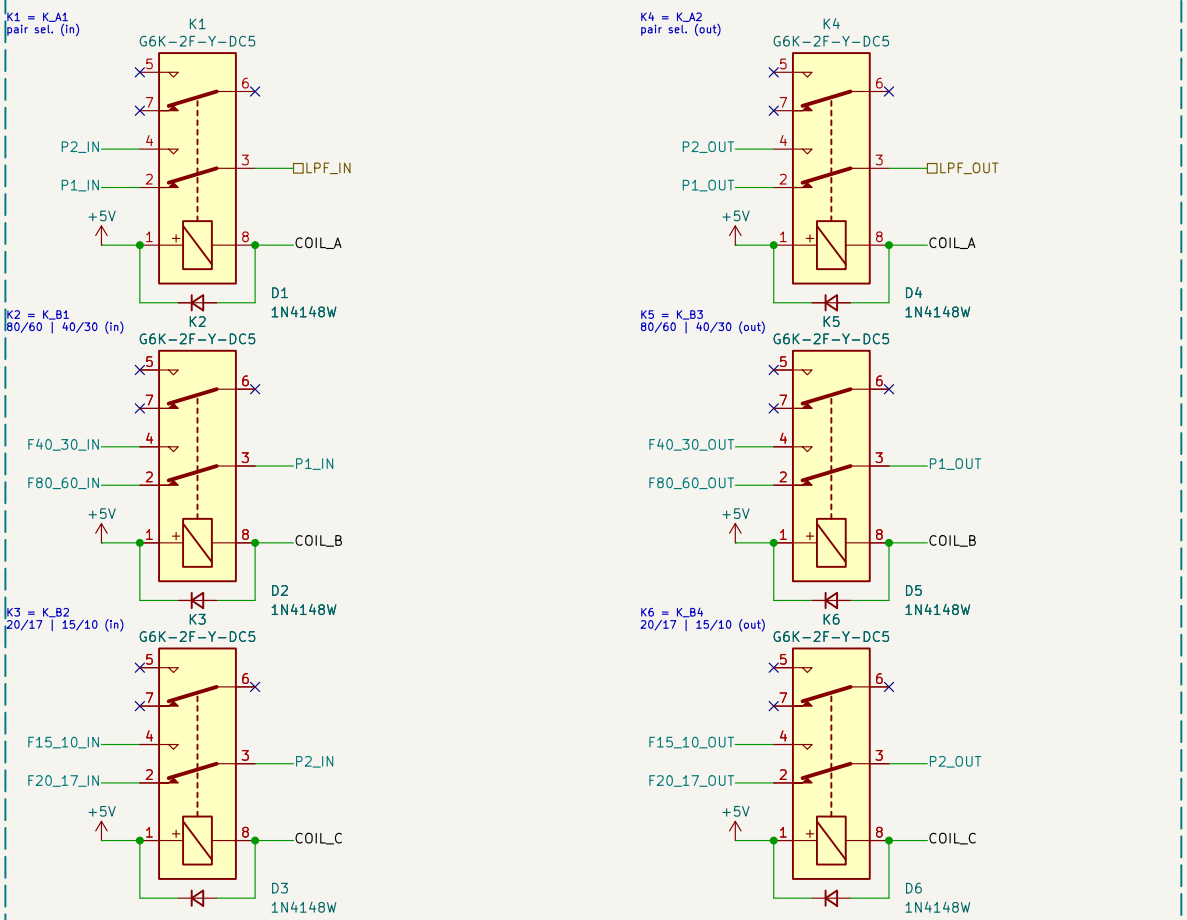
Size: A4	Date: 2026-10-25	**Rev: C**
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F2–F5 low-pass filters (T30 toroids)

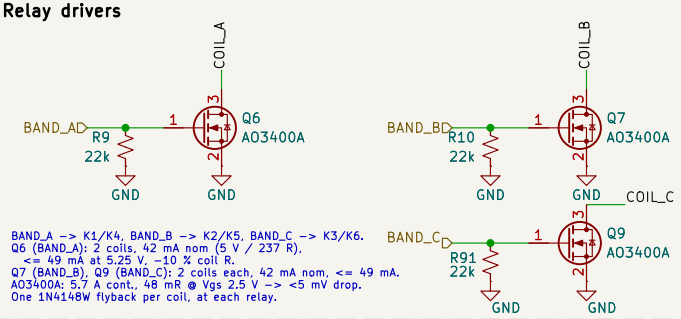


Four 7th-order Cauer LPFs, hand-wound T30 toroids.
Schematic + winding table on sheet LPF Filters.
Designed for 1 W output max.

Relay tree (2 levels, input + output mirrored; all off = 80/60 m)



Relay drivers



BAND_A -> K1/K4, BAND_B -> K2/K5, BAND_C -> K3/K6.
Q6 (BAND_A): 2 coils, 42 mA nom (5 V / 237 R).
<= 49 mA at 5.25 V, -10 % coil R.
Q7 (BAND_B), Q9 (BAND_C): 2 coils each, 42 mA nom, <= 49 mA.
AO3400A: 5.7 A cont., 48 mR @ Vgs 2.5 V -> <5 mV drop.
One 1N4148W flyback per coil, at each relay.

Band code + notes

Band code (BAND_A, BAND_B, BAND_C):
0 0 0 80/60 m (all relays off, NC path, default)
0 1 0 40/30 m (K2, K5 on)
1 0 0 20/17 m (K1, K4 on)
1 0 1 15/12/10 m (K1, K4, K3, K6 on)
Other codes unused; never more than 4 coils on.
Unselected filters float (open at both ends).
+5V relay load: 0 / 42 / 42 / 84 mA nominal (<=98 mA worst).
Relay pins: COM right, NC lower left, NO upper left; pole 2 unused.
Input and output paths never share a relay body.
Layout: keep K1–K2–K3 and K4–K5–K6 traces short,
50 ohm CPWG over solid GND, filters next to their relays.

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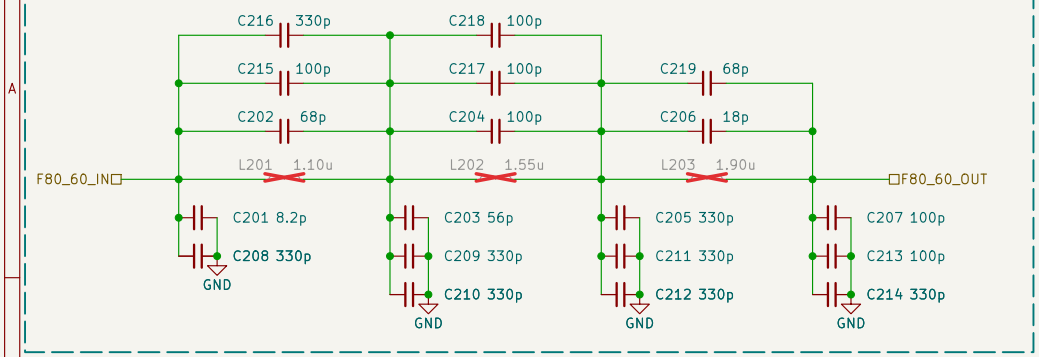
4 7th-order Cauer LPFs, 6-relay 2-level tree (BAND_A/B/C, 3-bit)

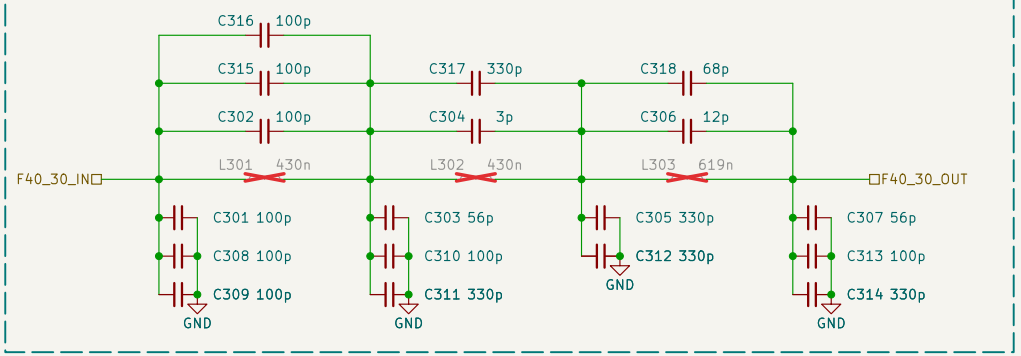
Etherkit

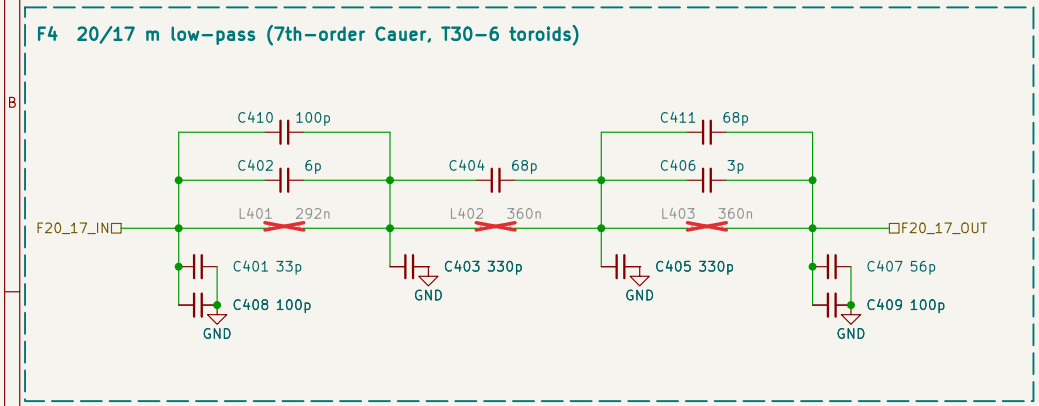
Sheet: /Filter Bank/
File: filter_bank.kicad_sch

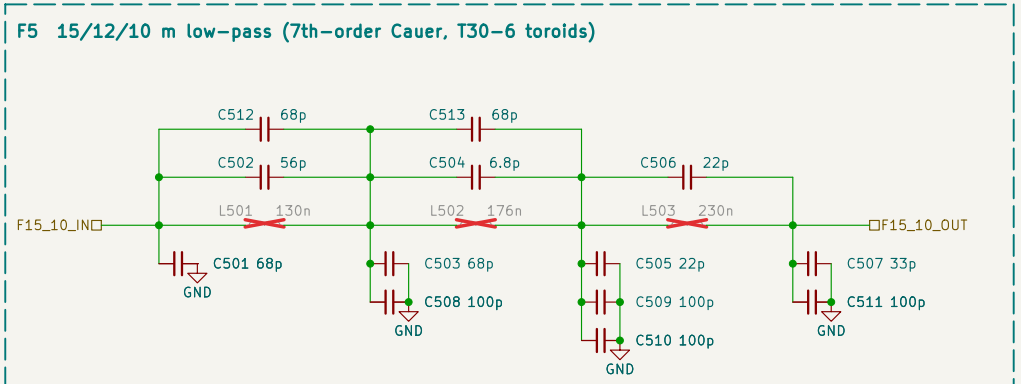
Title: OpenBeacon 2 – Low-Pass Filter Bank

Size: A4	Date: 2026-10-25	Rev: C
KiCad E.D.A. 10.0.7		Id: 3/6









Toroid winding table (hand-wound, not placed by JLC)

Ref	Filter / bands	Inductance	Core (Kits and Parts)	Turns	Wire
L201	F2 80/60 m	1.10 uH	T30-2 (red)	16	#28 AWG enam.
L202	F2 80/60 m	1.55 uH	T30-2 (red)	19	#28 AWG enam.
L203	F2 80/60 m	1.90 uH	T30-2 (red)	21	#30 AWG enam.
L301	F3 40/30 m	430 nH	T30-2 (red)	10	#24 AWG enam.
L302	F3 40/30 m	430 nH	T30-2 (red)	10	#24 AWG enam.
L303	F3 40/30 m	619 nH	T30-2 (red)	12	#26 AWG enam.
L401	F4 20/17 m	292 nH	T30-6 (yellow)	9	#24 AWG enam.
L402	F4 20/17 m	360 nH	T30-6 (yellow)	10	#24 AWG enam.
L403	F4 20/17 m	360 nH	T30-6 (yellow)	10	#24 AWG enam.
L501	F5 15/12/10 m	130 nH	T30-6 (yellow)	6	#24 AWG enam.
L502	F5 15/12/10 m	176 nH	T30-6 (yellow)	7	#24 AWG enam.
L503	F5 15/12/10 m	230 nH	T30-6 (yellow)	8	#24 AWG enam.

Cores: Micrometals T30-2 (red, F2/F3) and T30-6 (yellow, F4/F5) from Kits and Parts (kitsandparts.com).
Turns = passes through the core centre. AL 4.3 nH/t^2 (-2), 3.6 nH/t^2 (-6), +/-5%%.
F5 has few turns: tune each toroid by spreading or compressing turns. Mount toroids vertically.
Rated for 1 W output max. All filter caps 0603 C0G/NP0 50 V; parallel parts sum to the design value.

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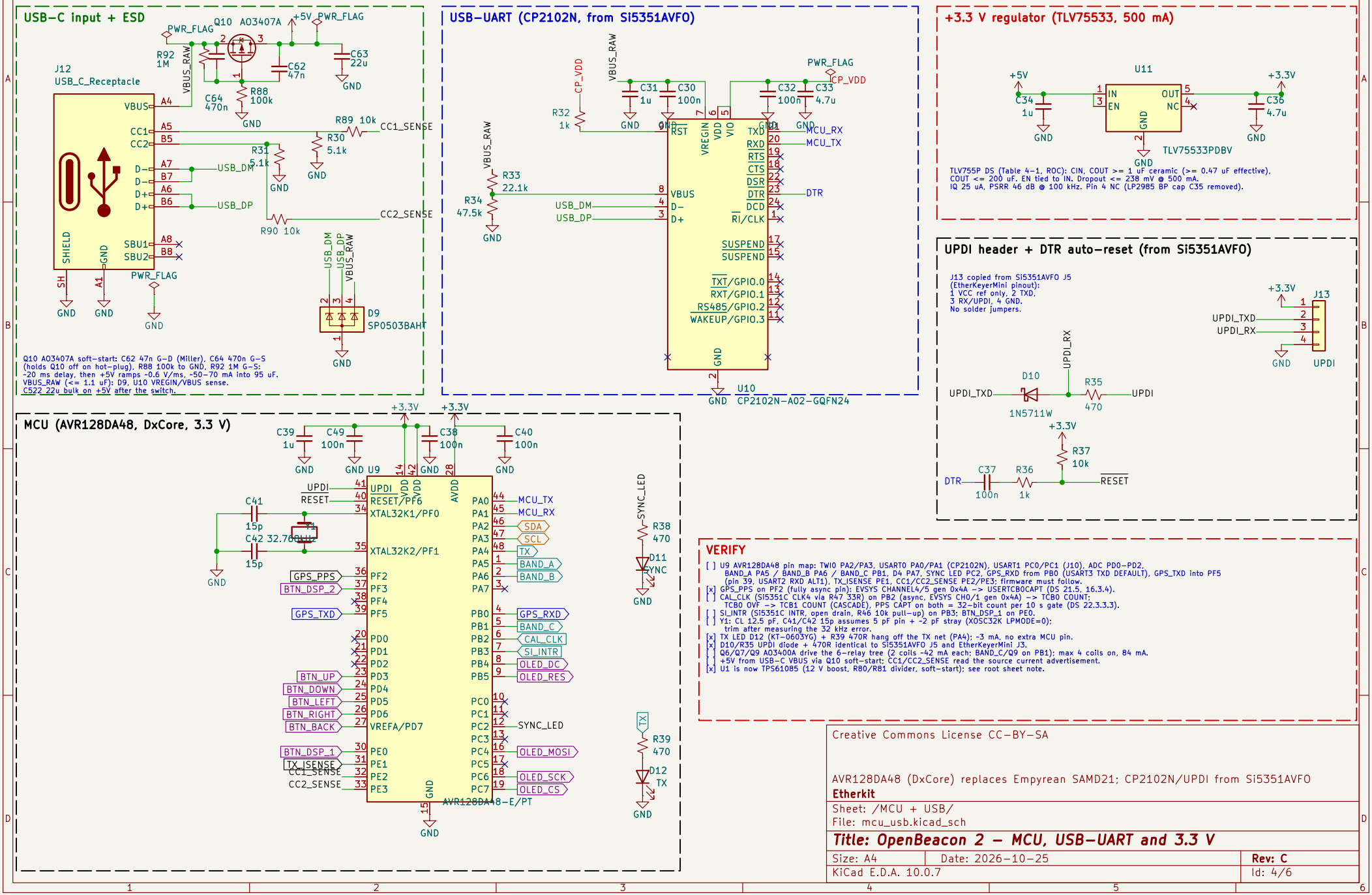
F2-F5 7th-order Cauer LPFs on hand-wound T30 toroids, 1 W max

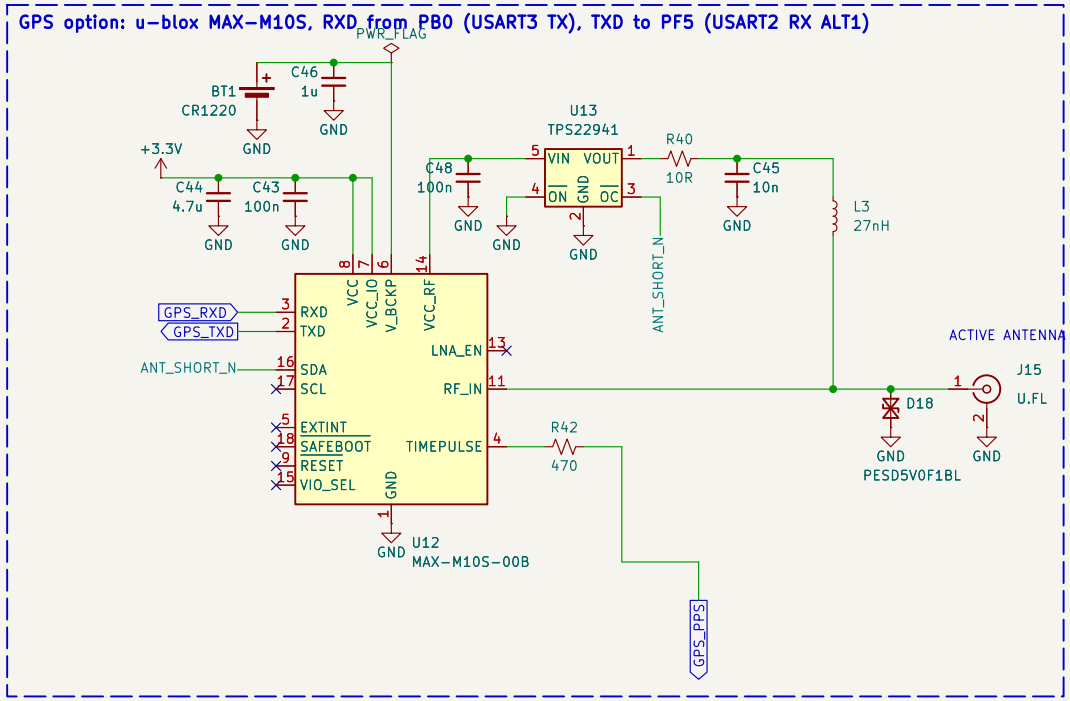
Etherkit

Sheet: /Filter Bank/LPF Filters/
File: lpf_filters.kicad_sch

Title: OpenBeacon 2 - LPFs F2-F5 (T30 toroids)

Size: A4	Date: 2026-10-25	**Rev: C**
KiCad E.D.A. 10.0.7		Id: 6/6


USB-C input + ESD
Q10 AO3407A
+5V_PWR_FLAG
PWR_FLAG
R92 1M
VBUS_RAW
C62 47n
C63 22u
C64 470n
R88 100k
J12
USB_C_Receptacle
VBUS
CC1
CC2
D-
D-
D+
D+
SBU1
SBU2
SHIELD
R89 10k
CC1_SENSE
R30 5.1k
R31 5.1k
USB_DM
USB_DP
R90 10k
CC2_SENSE
D9
SP0503BAHT
Q10 AO3407A soft-start: C62 47n G-D (Miller), C64 470n G-S (holds Q10 off on hot-plug), R88 100k to GND, R92 1M G-S: ~20 ms delay, then +5V ramps ~0.6 V/ms, ~50-70 mA into 95 uF. VBUS_RAW (<= 1.1 uF): D9, U10 VREGIN/VBUS sense. C522 22u bulk on +5V after the switch.
USB-UART (CP2102N, from Si5351AVFO)
CP_VDD
R32 1k
C31 1u
C30 100n
C32 100n
C33 4.7u
R33 22.1k
R34 47.5k
RST
VREGIN
VDD
VIO
VBUS
D-
D+
TXD
RXD
RTS
CTS
DSR
DTR
DCD
RI/CLK
SUSPEND
TXT/GPIO.0
RXT/GPIO.1
RS485/GPIO.2
WAKEUP/GPIO.3
MCU_RX
MCU_TX
DTR
U10
CP2102N-A02-GQFN24
+3.3 V regulator (TLV75533, 500 mA)
U11
C34 1u
C36 4.7u
IN
EN
OUT
NC
TLV75533PDBV
TLV755P DS (Table 4-1, ROC): CIN, COUT >= 1 uF ceramic (>= 0.47 uF effective), COUT <= 200 uF. EN tied to IN. Dropout <= 238 mV @ 500 mA. IQ 25 uA, PSRR 46 dB @ 100 kHz. Pin 4 NC (LP2985 BP cap C35 removed).
UPDI header + DTR auto-reset (from Si5351AVFO)
J13 copied from Si5351AVFO J5 (EtherKeyerMini pinout): 1 VCC ref only, 2 TXD, 3 RX/UPDI, 4 GND. No solder jumpers.
J13
UPDI_TXD
UPDI_RX
UPDI
D10
1N5711W
R35 470
R37 10k
C37 100n
R36 1k
DTR
RESET
MCU (AVR128DA48, DxCore, 3.3 V)
C39 1u
C49 100n
C38 100n
C40 100n
U9
UPDI
RESET/PF6
XTAL32K1/PF0
XTAL32K2/PF1
C41 15p
C42 32.768kHz
15p
PF2
PF3
PF4
PF5
PD0
PD1
PD2
PD3
PD4
PD5
PD6
VREFA/PD7
PE0
PE1
PE2
PE3
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PB0
PB1
PB2
PB3
PB4
PB5
PC0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
GPS_PPS
BTN_DSP_2
GPS_TXD
BTN_UP
BTN_DOWN
BTN_LEFT
BTN_RIGHT
BTN_BACK
BTN_DSP_1
TX_ISENSE
CC1_SENSE
CC2_SENSE
MCU_TX
MCU_RX
SDA
SCL
TX
BAND_A
BAND_B
GPS_RXD
BAND_C
CAL_CLK
SI_INTR
OLED_DC
OLED_RES
SYNC_LED
OLED_MOSI
OLED_SCK
OLED_CS
AVR128DA48-E/PT
R38 470
D11
SYNC
R39 470
D12
TX
VERIFY
[ ] U9 AVR128DA48 pin map: TWI0 PA2/PA3, USART0 PA0/PA1 (CP2102N), USART1 PC0/PC1 (J10), ADC PD0-PD2, BAND_A PA5 / BAND_B PA6 / BAND_C PB1, D4 PA7, SYNC LED PC2, GPS_RXD from PB0 (USART3 TXD DEFAULT), GPS_TXD into PF5 (pin 39, USART2 RXD ALT1), TX_ISENSE PE1, CC1/CC2_SENSE PE2/PE3; firmware must follow.
[x] GPS_PPS on PF2 (fully async pin): EVSYS CHANNEL4/5 gen 0x4A -> USERTCBOCAPT (DS 21.5, 16.3.4).
[ ] CAL_CLK (SI5351C CLK4 via R47 33R) on PB2 (async, EVSYS CH0/1 gen 0x4A) -> TCB0 COUNT; TCB0 OVF -> TCB1 COUNT (CASCADE), PPS CAPT on both = 32-bit count per 10 s gate (DS 22.3.3.3).
[ ] SI_INTR (SI5351C INTR, open drain, R46 10k pull-up) on PB3; BTN_DSP_1 on PE0.
[ ] Y1: CL 12.5 pF, C41/C42 15p assumes 5 pF pin + ~2 pF stray (XOSC32K LPMODE=0); trim after measuring the 32 kHz error.
[x] TX LED D12 (KT-0603YG) + R39 470R hang off the TX net (PA4); ~3 mA, no extra MCU pin.
[x] D10/R35 UPDI diode + 470R identical to Si5351AVFO J5 and EtherKeyerMini J3.
[ ] Q6/Q7/Q9 AO3400A drive the 6-relay tree (2 coils ~42 mA each; BAND_C/Q9 on PB1); max 4 coils on, 84 mA.
[ ] +5V from USB-C VBUS via Q10 soft-start; CC1/CC2_SENSE read the source current advertisement.
[x] U1 is now TPS61085 (12 V boost, R80/R81 divider, soft-start); see root sheet note.
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AVR128DA48 (DxCore) replaces Empyrean SAMD21; CP2102N/UPDI from Si5351AVFO
Etherkit
Sheet: /MCU + USB/
File: mcu_usb.kicad_sch
Title: OpenBeacon 2 - MCU, USB-UART and 3.3 V
Size: A4
Date: 2026-10-25
Rev: C
KiCad E.D.A. 10.0.7
Id: 4/6



VERIFY

- [x] Antenna short: TPS22941 limit 40–80 mA, auto-restart, avg <= 27 mA (was 320 mA).
- [x] V_BCKP: CR1220 in KH-BS1220-2-SMT (C5365932): C47/D13/R41 removed.
- [ ] BT1 footprint: draw from KH-BS1220-2-SMT DS.
- [ ] U13 TPS22941 JLC stock thin (26 pcs).
- [ ] Short peak 80 mA > VCC_RF 50 mA op. (DS T13) for <= 20 ms bursts; confirm on bench.
- [ ] V_IO ramp 25–35000 us/V (DS 4.1): measure.
- [ ] GPS_PPS MCU pin: see mcu_usb.

RECEIVER: VCC + V_IO tied to +3.3V; VIO_SEL open = 3.3 V I/O (IM 4.1.2).
C43/C44 beside pins 7/8, short GND vias; no series R > 0.2 ohm in VCC (IM 4.1.1).
RESET_N, EXTINT, SCL, SAFEBOOT_N, LNA_EN left open (DS Table 10).
SDA = ANT_SHORT_N (IM 3.3.1.2): CFG-HW-ANT_CFG_SHORTDET=1, ANT_SUP_SHORT_PIN=SDA PIO.
PPS: TIMEPULSE = SAFEBOOT_N via 1k inside (DS fn 15); nothing may hold it low at power-up, so no pull-down. R42 limits PPS fault current to ~7 mA (I_PIO 10 mA).

ANTENNA: RF_IN is DC-blocked inside the module (IM Table 1), no external DC block.
Bias-T VCC_RF -> U13 TPS22941 -> R40 10R -> C45 10n -> L3 27 nH onto RF line (IM 3.3.1.2, 4.3.4).
U13: I_LIM 40–80 mA, 5–20 ms blanking, 40–160 ms restart (TPS2294x DS 7.5): short avg <= 27 mA, typ 7 mA (VCC_RF abs max 250 mA, DS Table 12). 25 mA antenna drop ~0.27 V -> ~2.9 V at J15.
Layout: 50 ohm CPWG pin 11 -> J15, L3 pad on the line, >= 5 mm from digital.
BACKUP: BT1 CR1220 (3 V, 35 mAh, user-fitted, NOT rechargeable, no charger) on V_BCKP;
C46 1u at pin 6 (IM 4.1.3). 35 mAh / 28 uA (DS Table 17) = ~1250 h (~52 days) powered off.

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MAX-M10S: RXD from PB0 (USART3 TX), TXD to PF5 (USART2 RX ALT1), PPS to J14, active-antenna b

Etherkit

Sheet: /GPS/
File: gps.kicad_sch

Title: OpenBeacon 2 – GPS (u-blox MAX-M10S)

Size: A4	Date: 2026-10-25	**Rev: C**
KiCad E.D.A. 10.0.7		Id: 5/6